计算性设计：人工智能赋能的建筑学转型

2024 中国建筑学会计算性设计学术论坛工作营成果展

Multi-Robot Collaborative Autonomous Construction of Reciprocal Shells 多机协作网壳搭建

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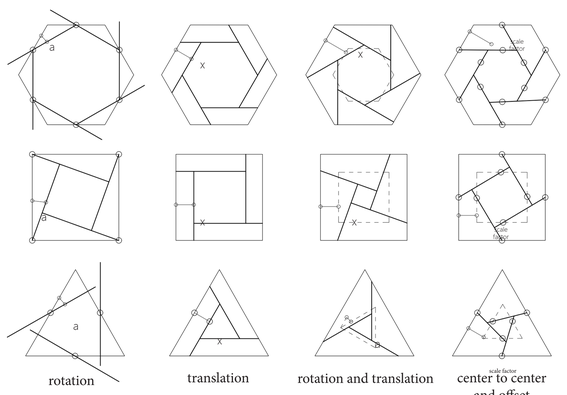
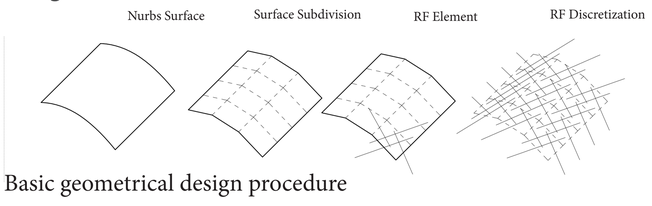
CONPONENT DESIGN | 节点设计

Basic Pattern



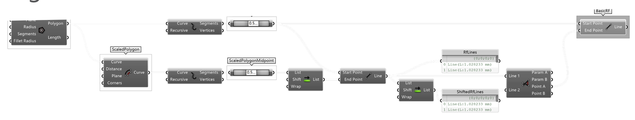
Geometrical RF design method applied at three different polygonal shape: triangle, quadrilateral, hexagon

Design Generation

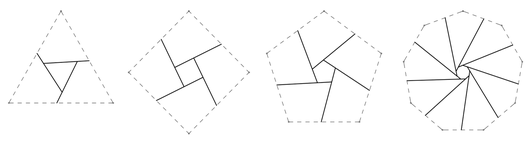


Basic geometrical design procedure

Single--unit Variation



Basic GH components to create a single-unit concentric RF

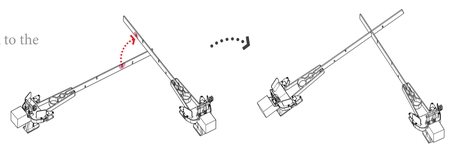


By increasing the number of the polygon edges eccentric single-unit RFs are created

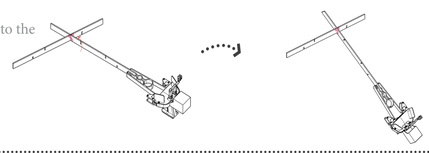
TASK UNDERSTANDING | 任务理解

SIMPLIFICATION

SEQUENCE 1
Pushing the lowerplank to the slot of the upper plank



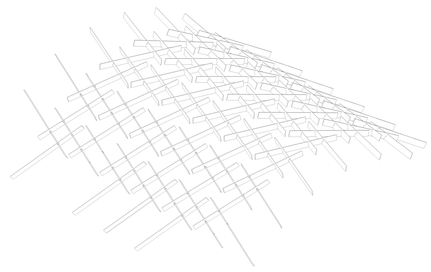
SEQUENCE 2
Pushing the upperplank to the slot of the lower plank



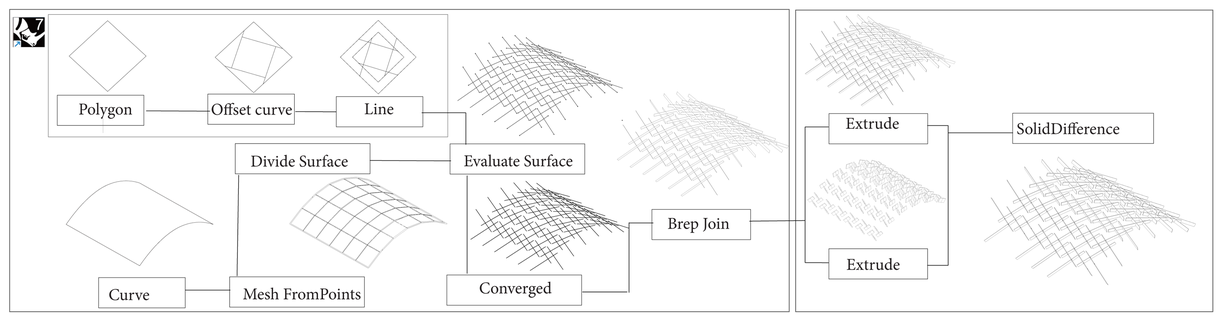
SEQUENCE 3
Rotate the plank to fit through thegap and rest between two planks

PARAMETRIC DESIGN | 参数化设计

ALIEN SURFACE

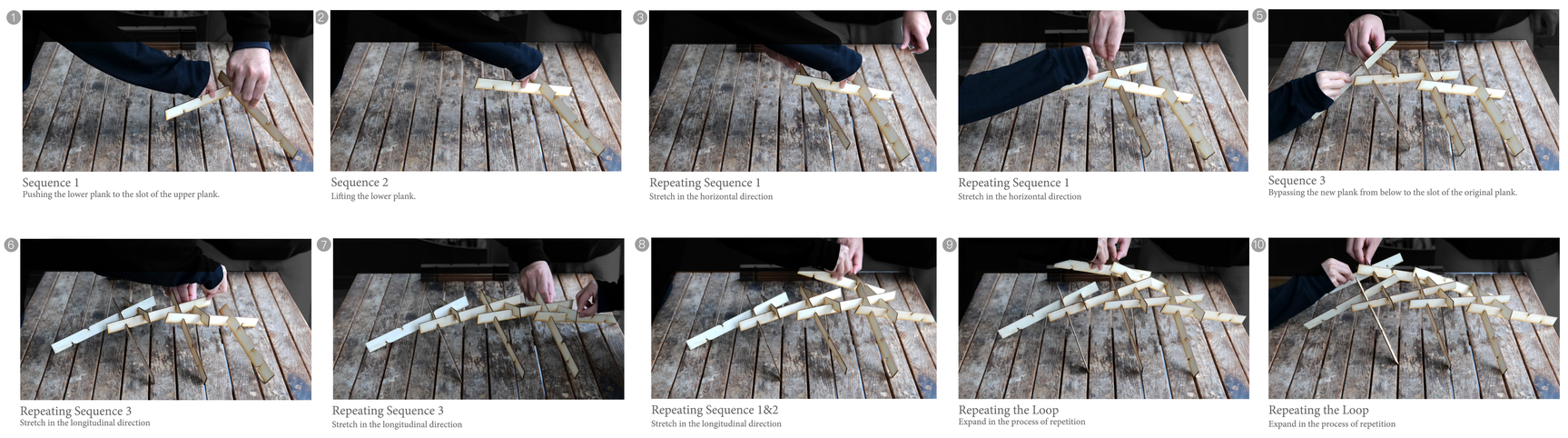


Simulation Process

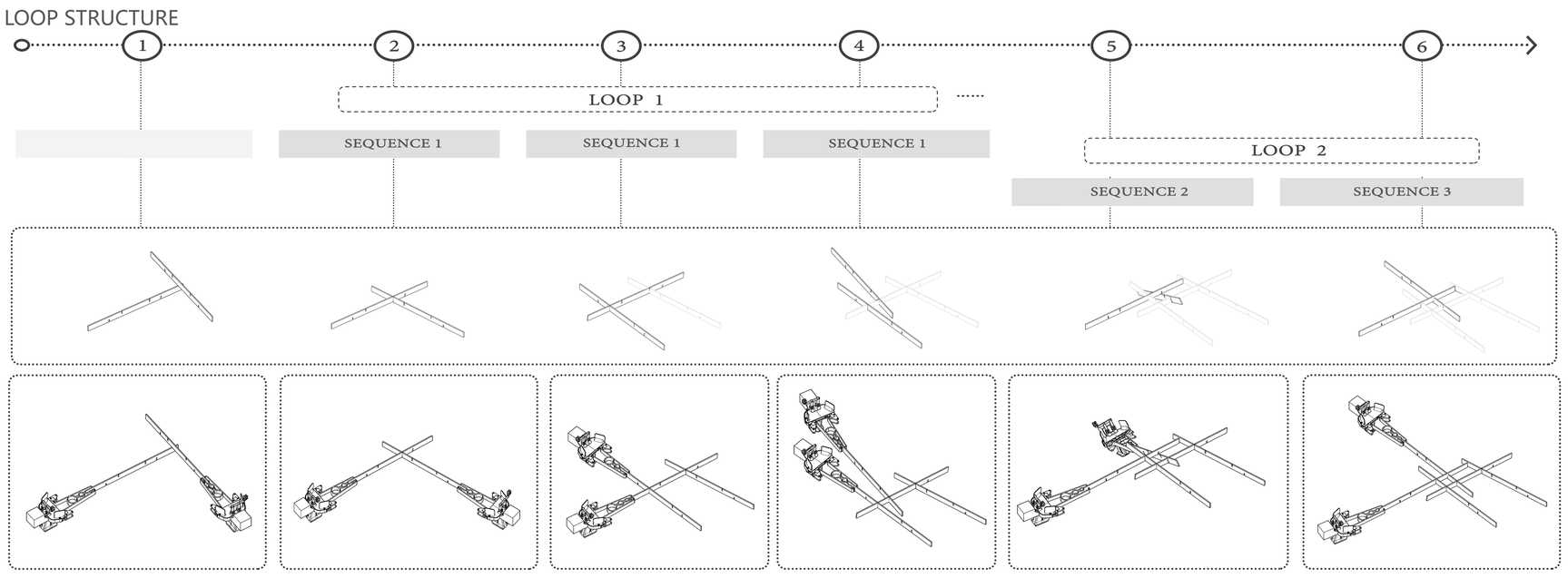


PHISICAL EXPLORATION | 搭建测试

Construction Process

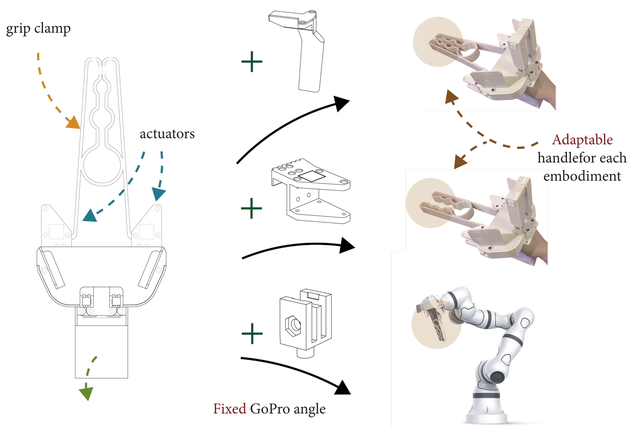


LOOP STRUCTURE



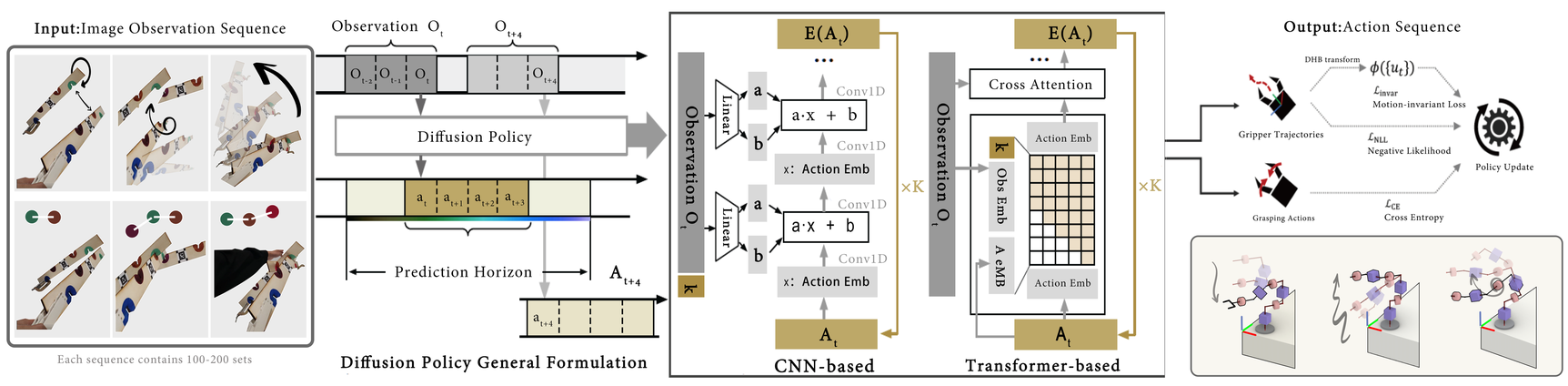
DATA COLLECTION | 数据采集

END EFFECTOR DESIGN



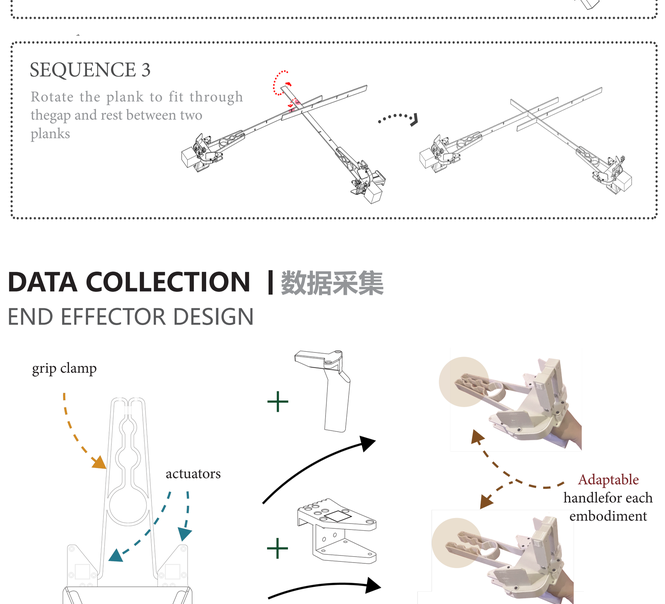
MACHINE LEARNING | 机器学习

NET STRUCTURE



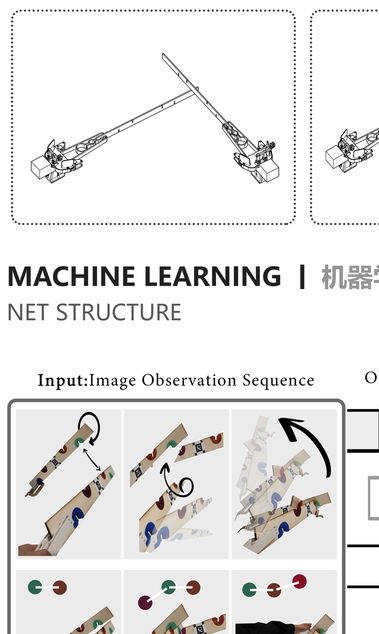
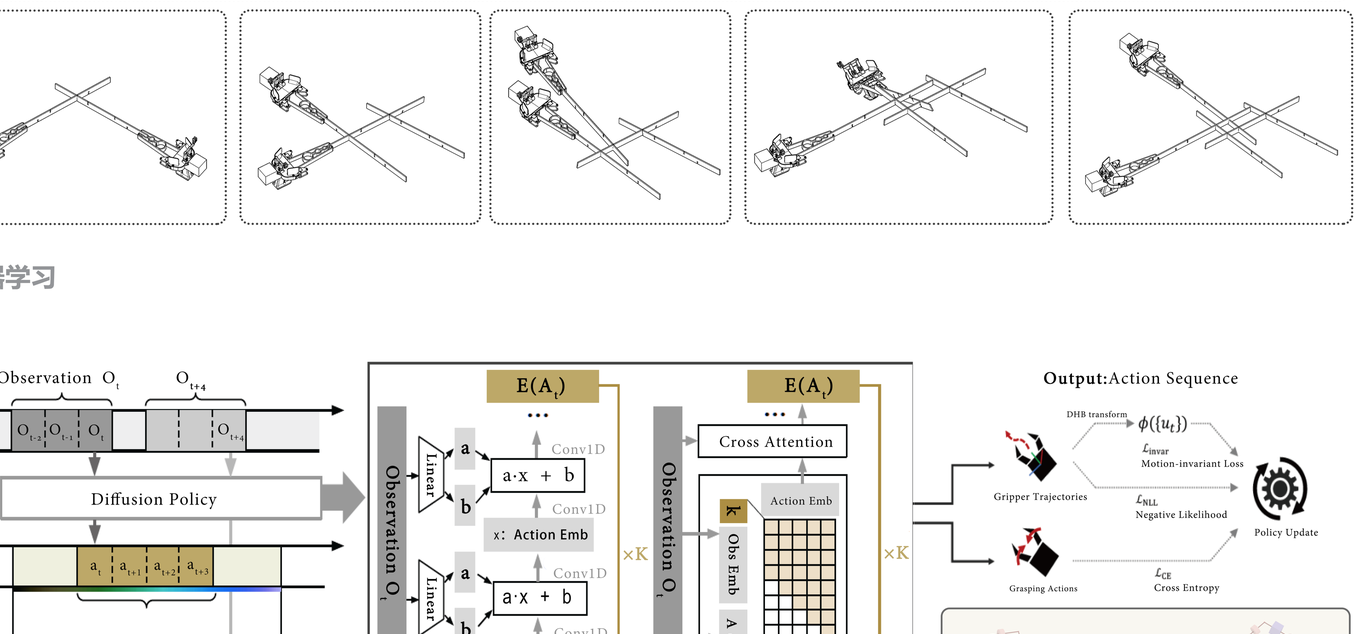
MULTI-PERSPECTIVE SHOOTING

Filmed with five horizontal angles and one top-down angle



Extract frames after collecting multiple sets of action videos

Extract frames to form our dataset



VIDEO FRAME EXTRACTION